THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 1 & 2: **Real Numbers, Polynomials** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $2^2 \times 5 \times 7$	$140 = 4 \times 35$
2	**(b)** 1	They have no common prime factor
3	**(a)** 720	$\frac{90 \times 144}{18} = 720$
4	**(c)** Incorrect; $\sqrt{9} = 3$ is rational	$\sqrt{9} = 3$
5	**(b)** Student A is correct	$3 \times 5^2 = 75$; B took the product
6	**(a)** $3, -5$	$(x-3)(x+5)$
7	**(b)** 4	Zeroes = x-intercepts
8	**(b)** 23	$8 + 15$
9	**(a)** $x^2 - 3x - 4$	Sum 3, product -4
10	**(d)** Incorrect — a non-zero constant has degree 0	$7 = 7x^0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$24 = 2^3 \times 3,\ 36 = 2^2 \times 3^2$; HCF $= 12$, LCM $= 72$; $12 \times 72 = 864 = 24 \times 36$ ✓	1+1
12	If $\frac{2\sqrt{3}}{5} = r$ (rational), then $\sqrt{3} = \frac{5r}{2}$ would be rational — contradiction.	2
13	$3(x-5)(x+5) \Rightarrow \pm 5$; sum $0 = -\frac{0}{3}$, product $-25 = \frac{-75}{3}$ ✓	1+1
14	$\frac{1}{2} + \frac{3}{2} + k = 0 \Rightarrow k = -2$; other zero $= -\frac{3}{2} - \frac{1}{2} = -2$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{11} = \frac{a}{b}$ (co-prime) $\Rightarrow 11b^2 = a^2 \Rightarrow 11 \mid a$; $a = 11c \Rightarrow b^2 = 11c^2 \Rightarrow 11 \mid b$. Contradiction.	1+1+1
16	$\sqrt{3}x^2 - 6x - 2x + 4\sqrt{3} = (x - 2\sqrt{3})(\sqrt{3}x - 2) \Rightarrow 2\sqrt{3},\ \frac{2}{\sqrt{3}}$. Sum $\frac{8}{\sqrt{3}}$ ✓, product $4 = \frac{4\sqrt{3}}{\sqrt{3}}$ ✓	1+1+1
17	$1001 = 7 \times 11 \times 13,\ 182 = 2 \times 7 \times 13$; HCF $= 91$ children; 11 biscuits and 2 chocolates each	2+1
	OR: $3x, 4x$ with HCF x: LCM $= 12x = 180 \Rightarrow x = 15 \Rightarrow$ 45 and 60	1+1+1
18	$\alpha + \beta = 3,\ \alpha\beta = 2$; new sum 7, product $2 + 2(3) + 4 = 12 \Rightarrow x^2 - 7x + 12$	1+1+1
	OR: $(\alpha+\beta)^2 - 2\alpha\beta = 64 - 2k = 40 \Rightarrow k = 12$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $312 = 2^3 \times 3 \times 13,\ 260 = 2^2 \times 5 \times 13$; HCF $= 52$ (ii) $6 + 5 = 11$ teams (iii) Yes, $208 = 52 \times 4$	2+2+1
	OR: (i) Standard contradiction proof. (ii) If $\sqrt{2} + \sqrt{3} = r$, then $\sqrt{3} = r - \sqrt{2} \Rightarrow 3 = r^2 - 2\sqrt{2}r + 2 \Rightarrow \sqrt{2} = \frac{r^2-1}{2r}$, rational — contradiction.	3+2
20	(i) 1 and 3 (ii) $x^2 - 4x + 3$ (iii) sum $4 = -\frac{-4}{1}$, product 3 ✓ (iv) $4 - 8 + 3 = -1$	1+1+2+1
	OR: $2x^2 - 3x - 9 = (2x+3)(x-3) \Rightarrow 3, -\frac{3}{2}$; sum $\frac{3}{2}$ ✓, product $-\frac{9}{2}$ ✓. Negatives $-3, \frac{3}{2} \Rightarrow 2x^2 + 3x - 9$	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapters 1 & 2: **Real Numbers, Polynomials** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $2^4 \times 3^3$	16×27
2	**(a)** 120	$2^3 \times 3 \times 5$
3	**(c)** π	π is irrational
4	**(b)** Incorrect — HCF ≤ LCM always	HCF divides LCM
5	**(a)** Student A is correct	72 is the LCM
6	**(d)** 2	$4 - 6 + 2 = 0$
7	**(b)** 3	$\frac{3}{k} = 1$
8	**(c)** downwards	$a < 0$
9	**(d)** $3, -7$	$(x-3)(x+7)$
10	**(b)** Student A is correct	$x^2 - 0x + (-5)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$40 = 2^3 \times 5,\ 36 = 2^2 \times 3^2,\ 126 = 2 \times 3^2 \times 7$; HCF $= 2$, LCM $= 2^3 \times 3^2 \times 5 \times 7 = 2520$	1+1
12	$12^n = 2^{2n} \times 3^n$ has no prime factor 5, so it cannot end in 0 or 5.	2
13	Sum 4, product $4 - 3 = 1 \Rightarrow x^2 - 4x + 1$	1+1
14	$\alpha + \beta = -\frac{5}{3},\ \alpha\beta = -\frac{2}{3}$; $\frac{25}{9} + \frac{4}{3} = \frac{37}{9}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	If $7 - 2\sqrt{3} = r$, then $\sqrt{3} = \frac{7-r}{2}$ is rational — contradiction.	1+1+1
16	$\frac{1}{6}(6x^2 + x - 12) = \frac{1}{6}(3x-4)(2x+3) \Rightarrow \frac{4}{3}, -\frac{3}{2}$; sum $-\frac{1}{6}$ ✓, product -2 ✓	1+1+1
17	LCM $= 2^3 \times 3^2 \times 5 = 360$; number $= 364$	1+1+1
	OR: HCF$(2048, 960)$: $2048 = 2^{11},\ 960 = 2^6 \times 15 \Rightarrow 64$	1+1+1
18	$x^2 - px - (p + c)$: $\alpha + \beta = p,\ \alpha\beta = -p - c$; $(\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1 = 1 - c$	1+1+1
	OR: $9 - 6 + k = 0 \Rightarrow k = -3$; other zero $= -2 - (-3) = 1$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) HCF$(120, 84) = 12$ m (ii) perimeter $408 \div 12 = 34$ poles (iii) $408 \div 6 = 68$	2+2+1
	OR: (i) $1176 = 2^3 \times 3 \times 7^2,\ 1260 = 2^2 \times 3^2 \times 5 \times 7$; HCF $= 84$, LCM $= 17640$; $84 \times 17640 = 1481760 = 1176 \times 1260$ ✓ (ii) $17640 = 15 \times 1176 = 14 \times 1260$	3+2
20	(i) $(x-2)(x-4) \Rightarrow x = 2$ and 4 (ii) $h(3) = -1 \Rightarrow$ 1 m below water (iii) sum 6, product 8 ✓	2+1+2
	OR: $p(2) = 8 - 20 + 4 + 8 = 0,\ p(-1) = -1 - 5 - 2 + 8 = 0,\ p(4) = 64 - 80 + 8 + 8 = 0$ ✓. Sum 5 ✓; $\Sigma\alpha\beta = -2 - 4 + 8 = 2$ ✓; product $-8 = -\frac{8}{1}$ ✓	2+3

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ANSWER KEY & MARKING SCHEME — SET C

Chapters 1 & 2: **Real Numbers, Polynomials** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 24	$2^3 \times 3$
2	**(c)** 12	$2^2 \times 3$
3	**(b)** 36	$2^2 \times 3^2$
4	**(a)** Incorrect — $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ is irrational	Numerator/denominator must be integers
5	**(b)** Student A is correct	It fails for three numbers in general
6	**(c)** ± 1	$(x-1)(x+1)$
7	**(c)** 2	$4 + 2a - 8 = 0$
8	**(b)** $4x^2 + x + 1$	$k(x^2 + \frac{1}{4}x + \frac{1}{4})$
9	**(c)** $-2, 5$	x-coordinates of intercepts
10	**(d)** Incorrect — $x^2 + 4 > 0$, so it has no real zero	$x^2 + 4 \geq 4$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$867 = 3 \times 17^2,\ 255 = 3 \times 5 \times 17 \Rightarrow$ HCF $= 51$	1+1
12	(i) $= 3$, rational (ii) $= 5 + 2\sqrt{6}$, irrational	1+1
13	$(x - \sqrt{3})(x - 1) \Rightarrow \sqrt{3}$ and 1	2
14	$\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2}{1} = 7$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Assume $\sqrt{13} = \frac{a}{b}$ (co-prime) $\Rightarrow 13b^2 = a^2 \Rightarrow 13 \mid a$; $a = 13c \Rightarrow b^2 = 13c^2 \Rightarrow 13 \mid b$. Contradiction.	1+1+1
16	Zeroes of $x^2 - 5x + 6$: 2, 3 $\Rightarrow$ new zeroes 4, 6 $\Rightarrow x^2 - 10x + 24 \Rightarrow p = -10,\ q = 24$	1+1+1
17	$\frac{2079 \times 27}{189} = 297$	1+1+1
	OR: LCM$(5, 6, 8) = 120 \Rightarrow 123$	1+1+1
18	$4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} = (4x - \sqrt{3})(\sqrt{3}x + 2) \Rightarrow \frac{\sqrt{3}}{4}, -\frac{2}{\sqrt{3}}$; sum $-\frac{5}{4\sqrt{3}}$ ✓, product $-\frac{1}{2}$ ✓	1+1+1
	OR: $a + b = 1$ and $4a - 2b = 10 \Rightarrow a = 2,\ b = -1$; product of zeroes $= 2 \Rightarrow$ third zero $= -1$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2 \times 3 \times 5,\ 3^2 \times 5,\ 2^2 \times 3 \times 5$ (ii) LCM $= 180$ min $\Rightarrow$ 9:00 a.m. (iii) 6, 9, 12, 3, 6, 9 $\Rightarrow$ 6 times	1+2+2
	OR: Contradiction proof with $3 \mid a,\ 3 \mid b$ (3). If $5 + 2\sqrt{3} = r$, $\sqrt{3} = \frac{r-5}{2}$ rational — contradiction (2)	3+2
20	(i) -2 and 4 (ii) $-(x+2)(x-4) = -x^2 + 2x + 8 \Rightarrow b = 2,\ c = 8$ (iii) sum $2 = -\frac{2}{-1}$, product $-8 = \frac{8}{-1}$ ✓ (iv) 9	1+2+1+1
	OR: $\alpha + \beta = \frac{5}{2},\ \alpha\beta = 1$. (i) $\frac{17}{4}$ (ii) $\frac{9}{4}$ (iii) $\frac{5}{2}$ (iv) $2x^2 - 5x + 2$ (zeroes 2 and $\frac{1}{2}$ are reciprocals)	1+1+1+2

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ANSWER KEY & MARKING SCHEME — SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $17 \times 19 \times 23$	$17 \times 19 = 323$, $323 \times 23 = 7429$
2	**(b)** 323	Distinct primes
3	**(b)** 216	12×18
4	**(c)** Incorrect — the factorisation is unique (FTA)	Fundamental Theorem of Arithmetic
5	**(b)** Student A is correct	$\sqrt{4 \times 3}$
6	**(c)** $x^2 - 4x$	$x(x-4)$
7	**(d)** −16	$\alpha\beta(\alpha+\beta) = -8 \times 2$
8	**(b)** $0, 2, -2$	$x(x-2)(x+2)$
9	**(b)** $a = c$	Product $\frac{c}{a} = 1$
10	**(a)** Incorrect — it has exactly one zero	$ax + b = 0 \Rightarrow x = -\frac{b}{a}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No common prime factor ⇒ HCF $= 1$; LCM $= 2^3 \times 3^2 \times 5^2 = 1800$	1+1
12	$\frac{306 \times 657}{22338} = 9$	2
13	$x^2 - 0x + \sqrt{5} = x^2 + \sqrt{5}$	2
14	$4 + 2 - 6 = 0$, $9 - 3 - 6 = 0$ ⇒ both are zeroes; sum $1 = -\frac{-1}{1}$ ✓	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	If $2 + 5\sqrt{3} = r$, then $\sqrt{3} = \frac{r-2}{5}$ is rational — contradiction.	1+1+1
16	$(v + 5\sqrt{3})(v - \sqrt{3}) \Rightarrow \sqrt{3}, -5\sqrt{3}$; sum $-4\sqrt{3}$ ✓, product -15 ✓	1+1+1
17	HCF$(825, 675, 450)$: $3 \times 5^2 \times 11$, $3^3 \times 5^2$, $2 \times 3^2 \times 5^2$ ⇒ 75 cm	1+1+1
	OR: $96 = 2^5 \times 3$, $360 = 2^3 \times 3^2 \times 5$; HCF $= 24$, LCM $= 1440$	1+1+1
18	Product $= 1$: $6a = a^2 + 9 \Rightarrow (a-3)^2 = 0 \Rightarrow a = 3$	1+1+1
	OR: $\alpha - \beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{49 - 40} = 3$; zeroes $-2, -5$ ✓	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $336 = 2^4 \times 3 \times 7$, $240 = 2^4 \times 3 \times 5$; HCF $= 48$ (ii) $7 + 5 = 12$ stacks (iii) 96 cm	2+2+1
	OR: (i) $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$; HCF 24, LCM 360 (ii) 15 (iii) No: $540 \div 24$ is not an integer	3+1+1
20	(i) $-(x-3)(x-7)$ ⇒ 3 and 7 (ii) $-25 + 50 - 21 = 4$ ⇒ ₹4000 (iii) sum $10 = -\frac{10}{-1}$, product $21 = \frac{-21}{-1}$ ✓	2+1+2
	OR: Product $= -12$: $(\sqrt{3})(-\sqrt{3})\gamma = -3\gamma = -12 \Rightarrow \gamma = 4$ (also sum $= 4$ ✓). $\Sigma\alpha\beta = -3 + 4\sqrt{3} - 4\sqrt{3} = -3 = \frac{-3}{1}$ ✓	3+2

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